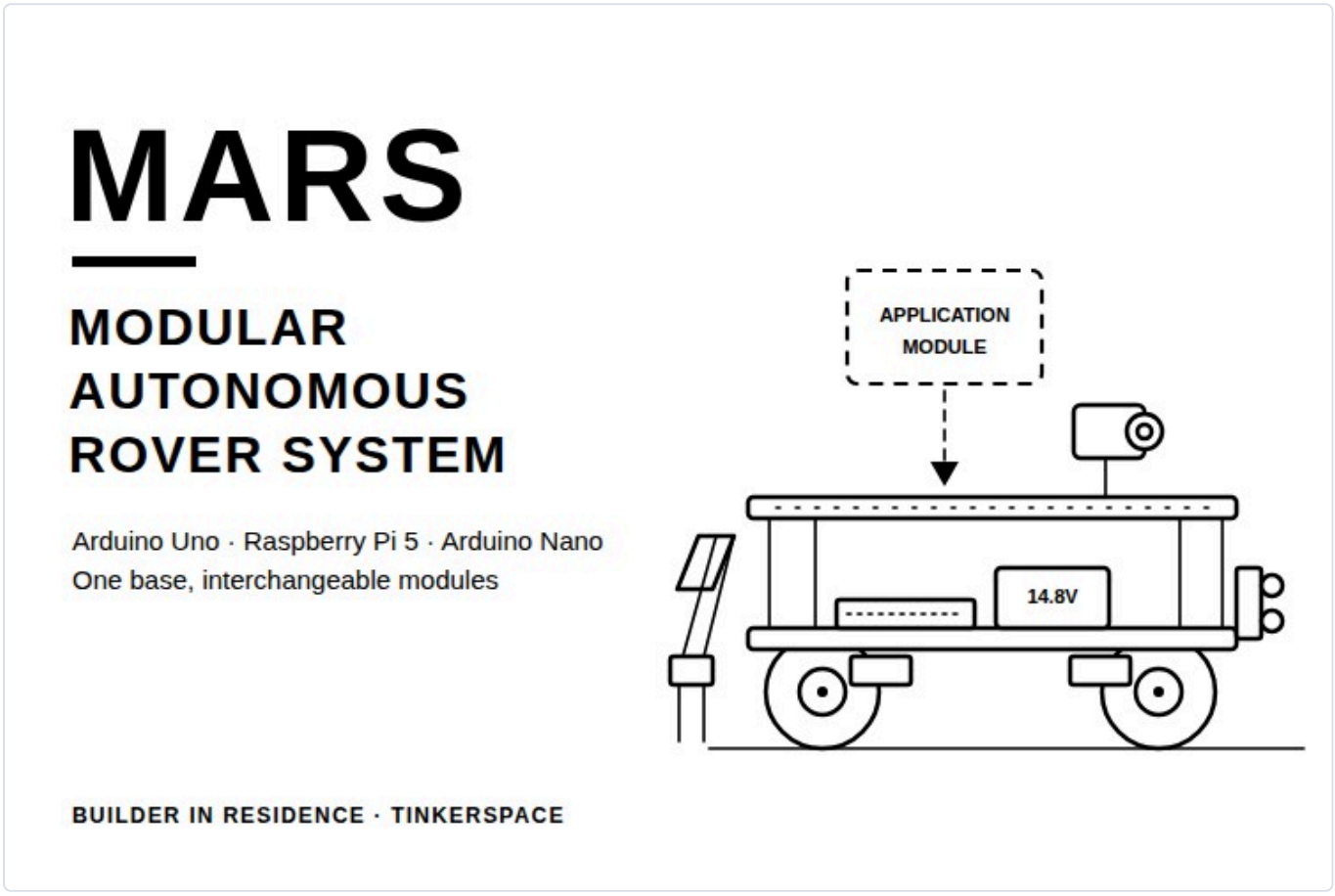
Draft for review **8 TODO** values still open

How to Build MARS, a Modular Rover Using Arduino Uno and Raspberry Pi 5

Draft article · MARS, built at Tinkerspace for Builder in Residence

Publishing details

Slug	`mars-modular-rover-arduino-uno-raspberry-pi-5`
Meta description	Build a modular rover with an Arduino Uno, HC-05 and three HC-SR04s, plus swappable Pi 5 camera and Arduino Nano sensor modules. Parts, wiring, STLs and tests. (159 characters)
Keywords	Arduino Uno · Raspberry Pi 5 · Arduino Nano · Modular Rover · HC-05 Bluetooth · HC-SR04 · L298N · Obstacle Detection · MQ Gas Sensor · DHT11 · Soil Moisture Sensor · 3D Printing · Fusion 360



Most student rovers are built for one job. A delivery rover needs something to carry the load, an inspection rover needs a camera, and a search rover needs more sensors again. So, could one rover base carry whichever module a job needs and swap it for the next job? That question became MARS, the Modular Autonomous Rover System.

In this project, we show how to build MARS, a modular rover with a 3D-printed two-deck chassis. The base rover runs on an Arduino Uno with an L298N motor driver, takes commands from a phone over an HC-05 Bluetooth module, and stops for obstacles using three HC-SR04 ultrasonic sensors at the front. On top of it sit two modules. The camera module streams live video from a USB webcam through a Raspberry Pi 5, and the environmental module uses an Arduino Nano with a gas sensor, a DHT11 and a soil-moisture probe on a servo lift. The modular rover drives and stops for obstacles, and both modules work, but each module still runs on its own controller, mapping and full autonomy were not reached, and the code for all three controllers is not in the repository yet. You can also check out similar robotics projects and Arduino projects done previously here at Circuit Digest.

> **Quick Answer: What is MARS?**
>
> **MARS** is a **modular rover** with a 3D-printed two-deck chassis. The base rover uses an **Arduino Uno**, an **L298N**, an **HC-05** and three **HC-SR04** sensors for Bluetooth driving and obstacle stops, and the upper deck takes swappable modules: a **Raspberry Pi 5** camera module for live video and an **Arduino Nano** environmental module with gas, temperature, humidity and soil sensors.

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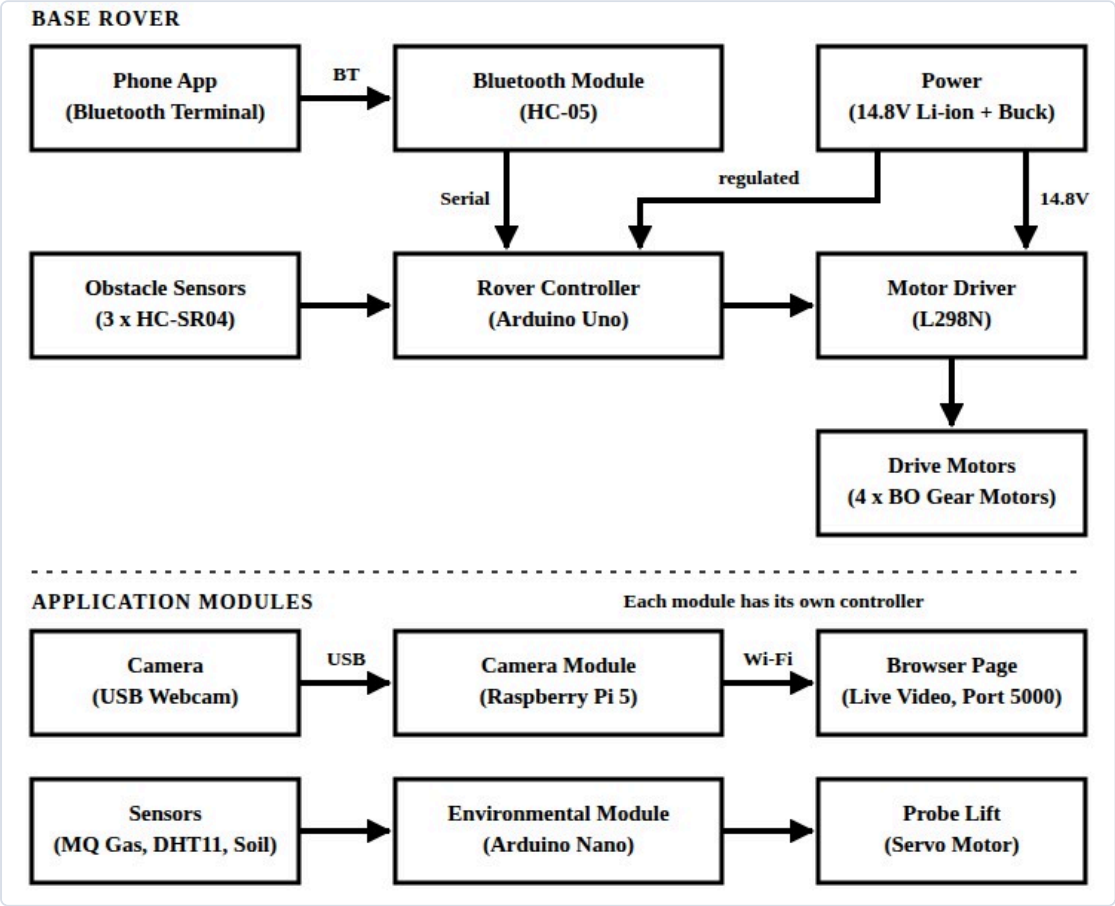
What Is MARS?

Disaster search, package delivery and remote inspection each need different hardware on a rover. Instead of a separate rover for every job, MARS uses one common base for movement, power and obstacle sensing, with an application module on top that changes with the task. To keep the build

achievable, it has two modules: a camera module and an environmental monitoring module. Circuit Digest has also built a Mars rover robot using Arduino with six BO motors and Bluetooth control. If interested, do check it out.

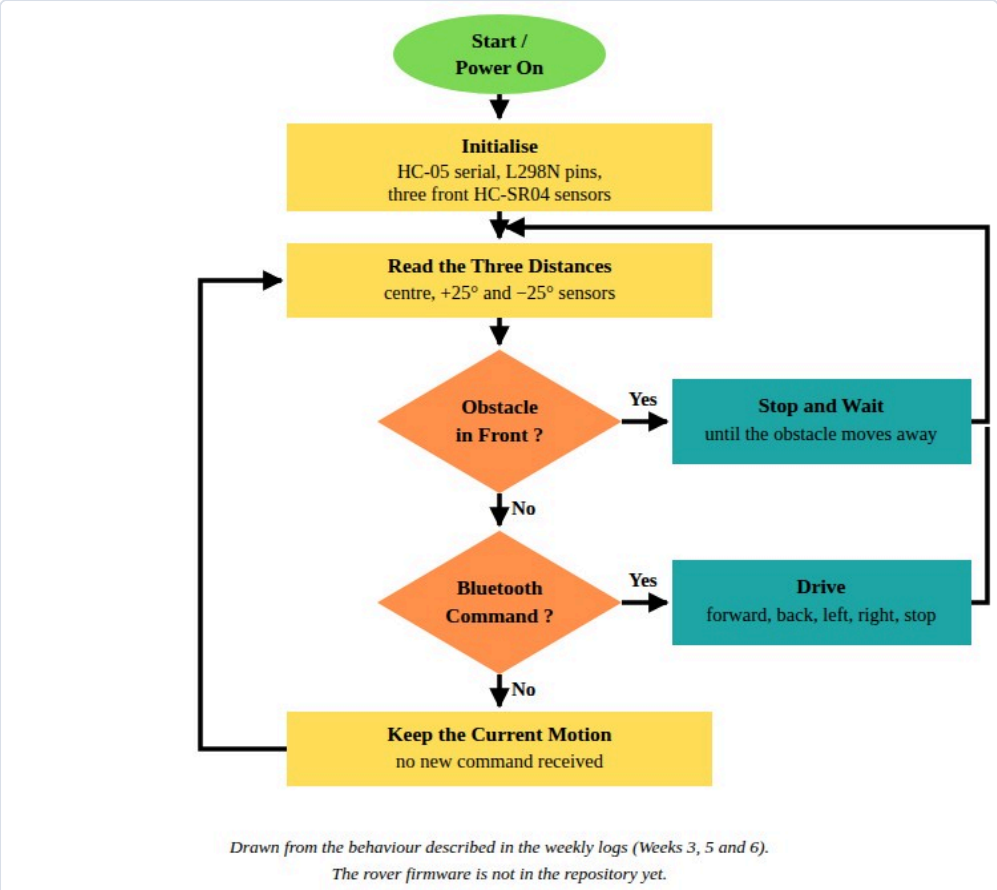
How Does the MARS Modular Rover Work?

The working is split between the base rover and the modules. On the base, a Bluetooth serial terminal app on the phone sends movement commands to the HC-05, which passes them to the Arduino Uno over serial, and the Uno drives four BO gear motors through the L298N. The three HC-SR04 sensors face forward, one at the centre and the other two at about +25° and −25°. When they see an obstacle, the rover stops and waits until it moves away. Below is the block diagram of the MARS modular rover.



The two modules sit on the upper deck and run on their own controllers. The camera module streams video from the webcam over Wi-Fi to a browser page, and the environmental module reads its sensors and moves the soil probe with a servo. There is no data link between a module and the base yet. Here is another web-controlled Raspberry Pi surveillance robotic car that pairs a Pi and a webcam with a rover.

Below is the flowchart of the base rover's control loop.

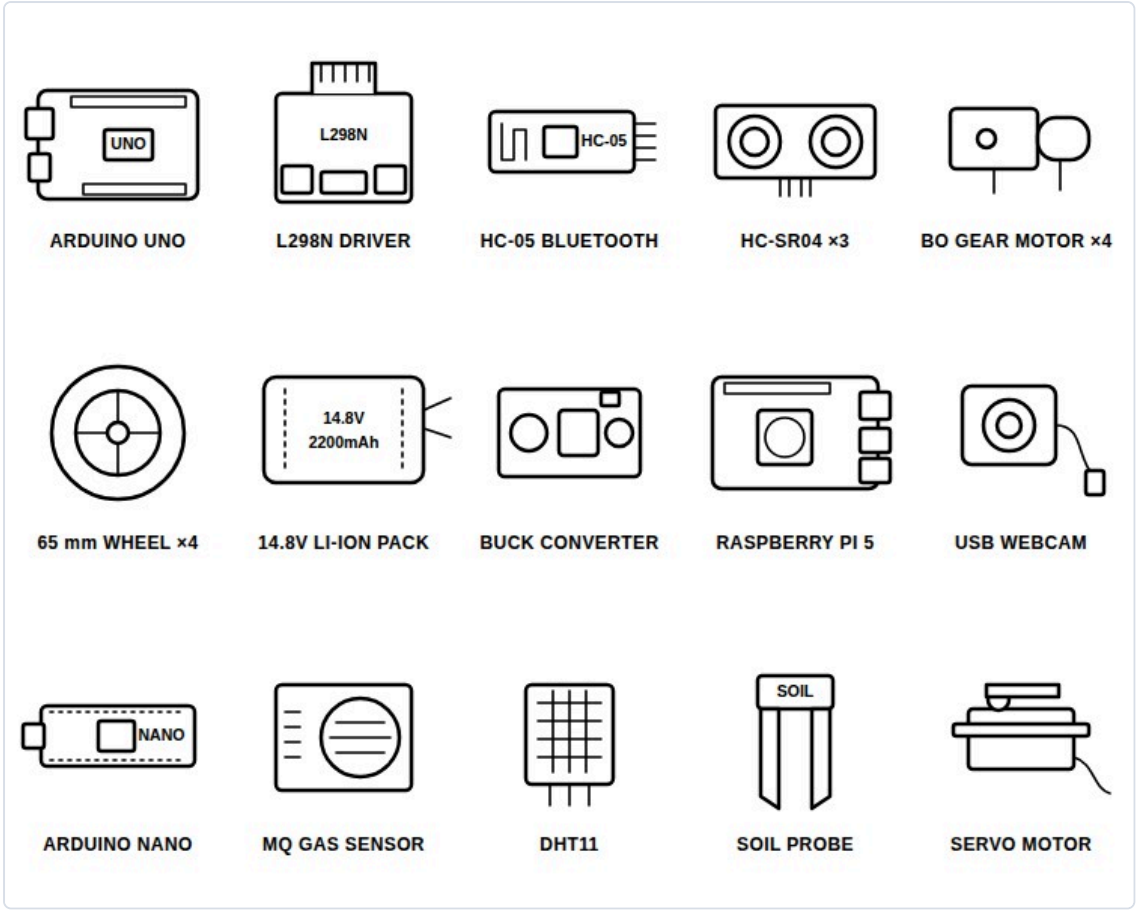


Drawn from the behaviour described in the weekly logs (Weeks 3, 5 and 6).
The rover firmware is not in the repository yet.

The rover firmware is not in the repository, so this flowchart follows the behaviour described in the build logs. Each pass reads the three distances first. An obstacle in front stops the rover until the path is clear, and otherwise the latest Bluetooth command sets the direction.

Components Required for the MARS Modular Rover

Below is the list of components used to build the base rover and the two modules.



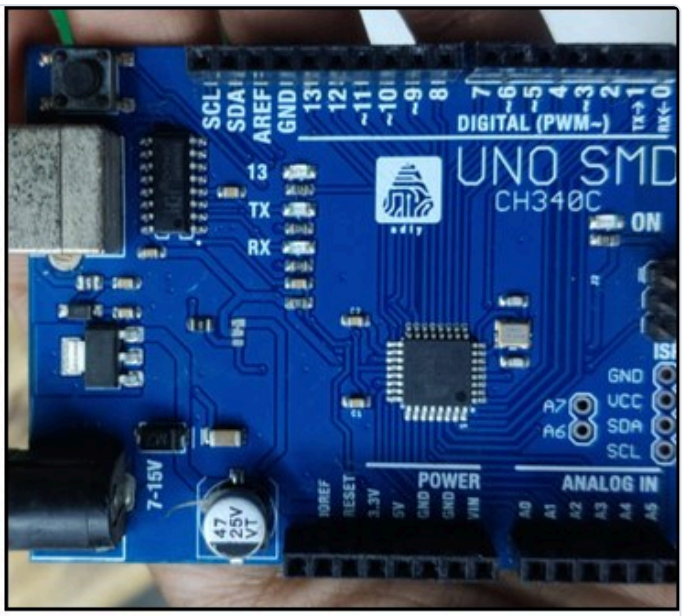
S.No	Component	Description	Quantity
1.	Arduino Uno	Base rover controller	1
2.	L298N Motor Driver	Drives the four motors	1
3.	HC-05 Bluetooth Module	Commands from the phone	1
4.	HC-SR04 Ultrasonic Sensor	Obstacle detection at the front	3
5.	BO Gear Motor TODO: voltage and RPM	Drive	4
6.	65 mm Universal Hub Tyre	Wheels	4
7.	14.8V 2200mAh Li-ion Pack	Main battery, 4S1P 18650	1
8.	Buck Converter	Supply for the electronics	1
9.	Perfboard and Wires	Permanent circuit	As required
10.	3D-Printed Decks and Lift Parts	Chassis and probe lift, five STL files	1 set
11.	Raspberry Pi 5 with Active Cooler	Camera module computer	1
12.	USB Webcam	Live video	1
13.	Arduino Nano	Environmental module controller	1
14.	MQ Gas Sensor TODO: MQ model	Gas detection	1

S.No	Component	Description	Quantity
15.	DHT11	Temperature and humidity	1
16.	Soil Moisture Probe	Soil moisture	1
17.	Servo Motor	Raises and lowers the probe	1

Below are the battery pack and the Uno board we used.



14.8V 2200mAh LI-ION PACK
4S1P, 18650 CELLS

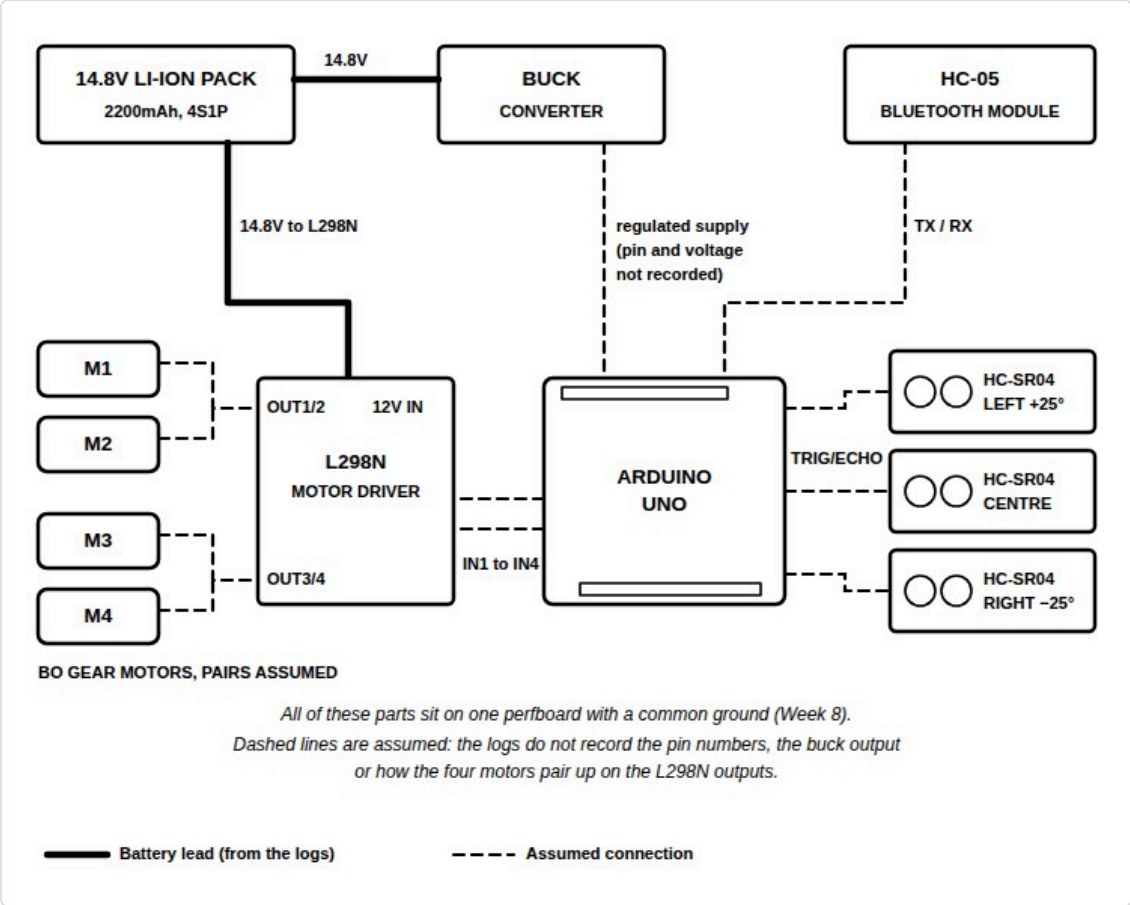


ARDUINO UNO (CH340C)
REPLACED THE ESP32-S3

The pack is a Wattnine 14.8V 2200mAh Li-ion pack, built as 4S1P from 18650 cells and rated for 12.2V to 16.8V. Our Arduino Uno is a CH340C-based UNO SMD board. We started on an ESP32-S3 with its own Wi-Fi control page, but it was damaged by a short circuit while the motor driver was being tested with the battery connected, so the build moved to the Uno and, since the Uno has no Wi-Fi, to an HC-05 for Bluetooth control. Use wheels made for your motors. The 65 mm Universal Hub tyres fit our motors better than the first wheels and made the rover more stable. The gas sensor is a blue MQ-series module, but its model is not marked in any of the photos. Apart from these parts, you need a 3D printer, a soldering iron and a multimeter. To learn how the L298N works, refer to our L298N motor driver with Arduino tutorial.

Wiring Diagram of the MARS Base Rover

Below is the wiring diagram of the MARS base rover.



The brain of the base rover is the Arduino Uno. The 14.8V Li-ion pack feeds the L298N's motor supply directly and also feeds the buck converter, which supplies the low-voltage electronics. Commands from the HC-05 reach the Uno over serial, and each HC-SR04 has a trigger and an echo line to the Uno. On the output side, the Uno drives the L298N's IN1 to IN4 inputs, and the four BO motors connect to the OUT1/OUT2 and OUT3/OUT4 terminals. All of this is soldered on one perfboard with a common ground. The build logs do not record the pin numbers, the buck converter's output voltage or which motors share an output, so those lines are dashed in the diagram. Choose your own pins and check each line with a multimeter before connecting the battery. That is all about the wiring. For three HC-SR04 sensors on one Arduino, the human following robot on Circuit Digest uses a similar arrangement.

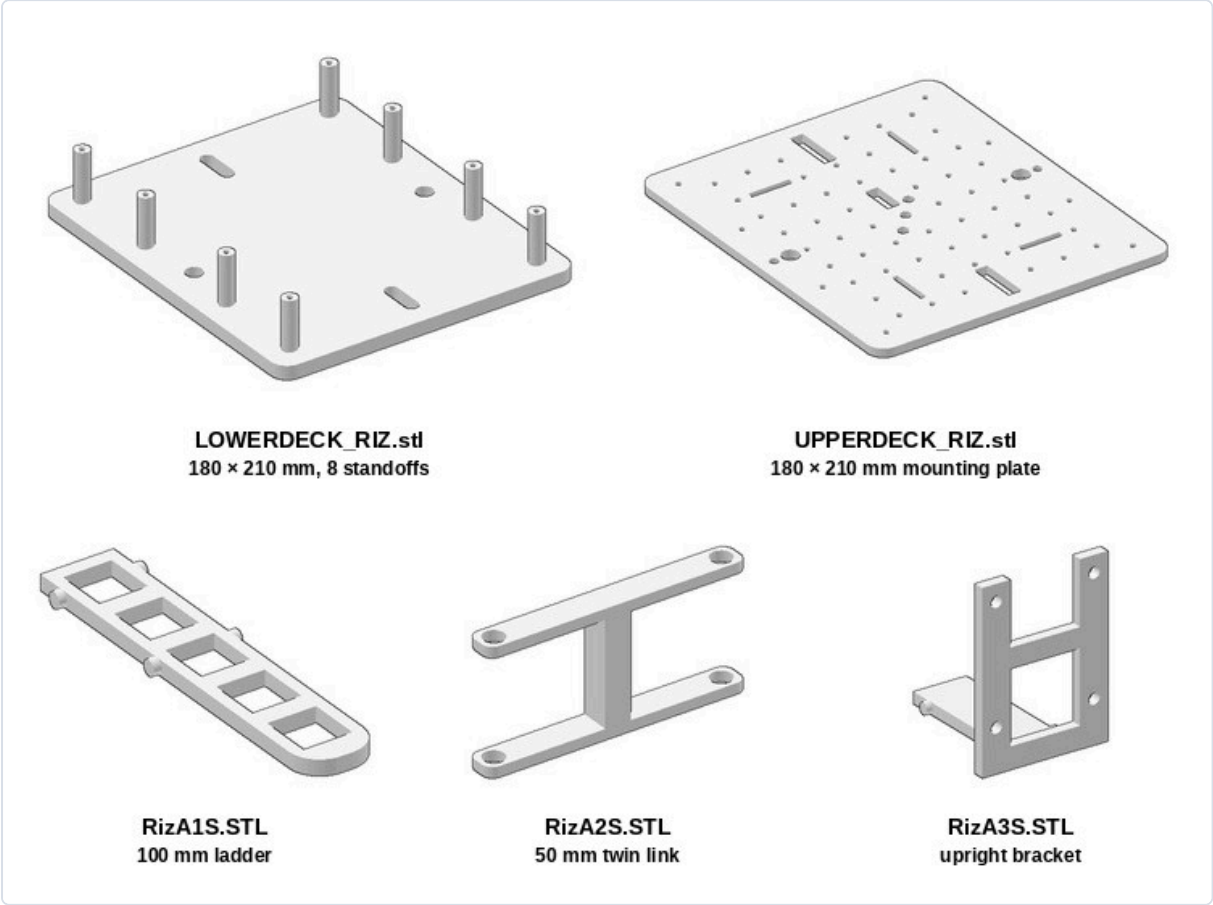
Connection Table

Connection	From	To	Recorded in the Repo?
Motor supply	14.8V pack	L298N motor supply input	Yes
Logic supply	14.8V pack through the buck converter	Arduino Uno and sensors	Yes, output voltage and pin not recorded
Bluetooth	HC-05 TX/RX	Arduino Uno serial pins	Assumed, pins not recorded
Obstacle sensors	Three HC-SR04 TRIG/ECHO	Arduino Uno digital pins	Assumed, pins not recorded

Connection	From	To	Recorded in the Repo?
Motor control	Arduino Uno	L298N IN1 to IN4	Assumed, pins not recorded
Motors	Four BO motors	L298N OUT1/OUT2 and OUT3/OUT4	Yes, pairing not recorded
Ground	Every board	Common ground on the perfboard	Yes

3D Printing the MARS Chassis and Probe Lift

The base was designed in Zoo Keeper and Fusion 360 at about 21 × 18 cm, with places for the battery holder, the ultrasonic sensor holders and a module area. Below are the printed parts, rendered from the STL files in the repository's cad folder.



The table below lists the five files and what each one is.

File in cad/	Part	Size
LOWERDECK_RIZ.stl	Lower deck with eight standoffs for the drive electronics	180 × 210 mm
UPPERDECK_RIZ.stl	Perforated upper deck where the modules mount	180 × 210 mm

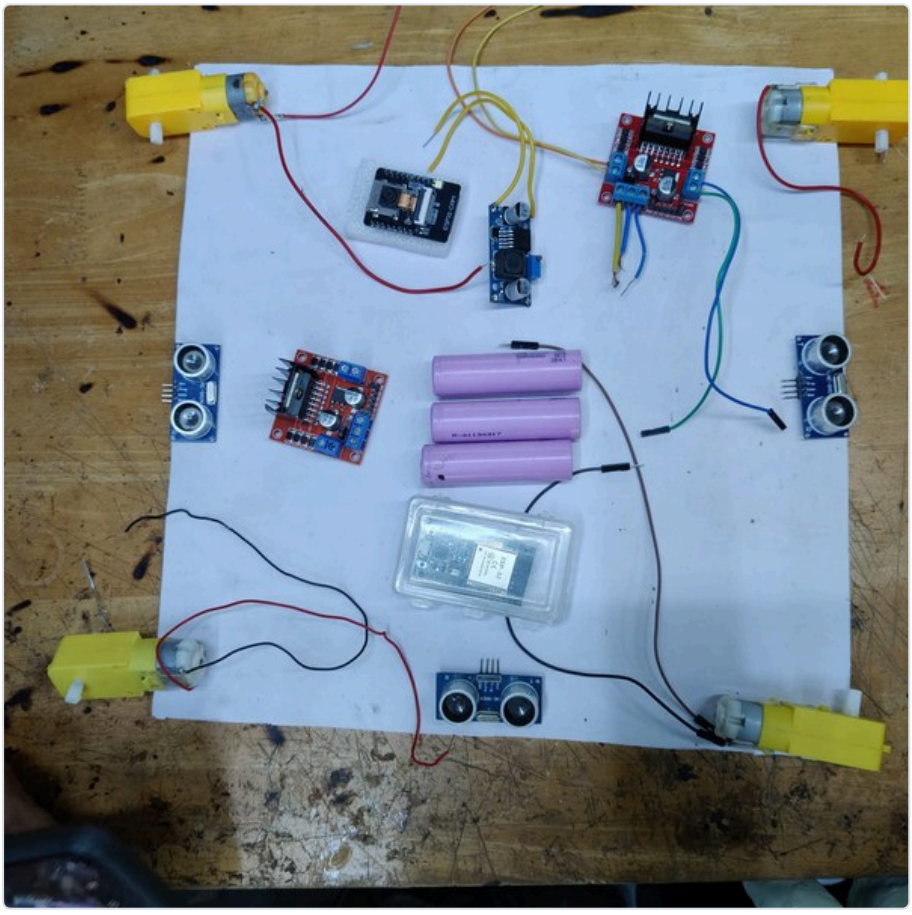
File in cad/	Part	Size
RizA1S.STL	Ladder that holds the soil probe	100 mm long
RizA2S.STL	Pair of links for the probe lift	50 mm
RizA3S.STL	Upright bracket for the probe lift	-

The roles of the three small parts are read from their shapes, as the repository does not name them. Print settings are not recorded in the repository **TODO: print material, layer height and infill**. Measure your own battery, sensors and boards before printing, as the printed size does not always match the CAD size. We corrected the deck dimensions a few times against the real parts before the final print. For more printed parts in real builds, take a look at the 3D printing projects on Circuit Digest.

Assembling the MARS Base Rover

Step 1⇒ Try the Layout on Foam Board

Before printing anything, lay the parts out on a 30 × 30 cm piece of foam board to find where the motors, the controller and the battery go. Foam board is cheap and easy to cut. Below is our first layout.



This layout still has the ESP32-S3 we started with. If you cannot find a holder for your battery, soldering its leads directly works as a temporary fix, but insulate every joint.

Step 2⇒ Test Each Part on Its Own

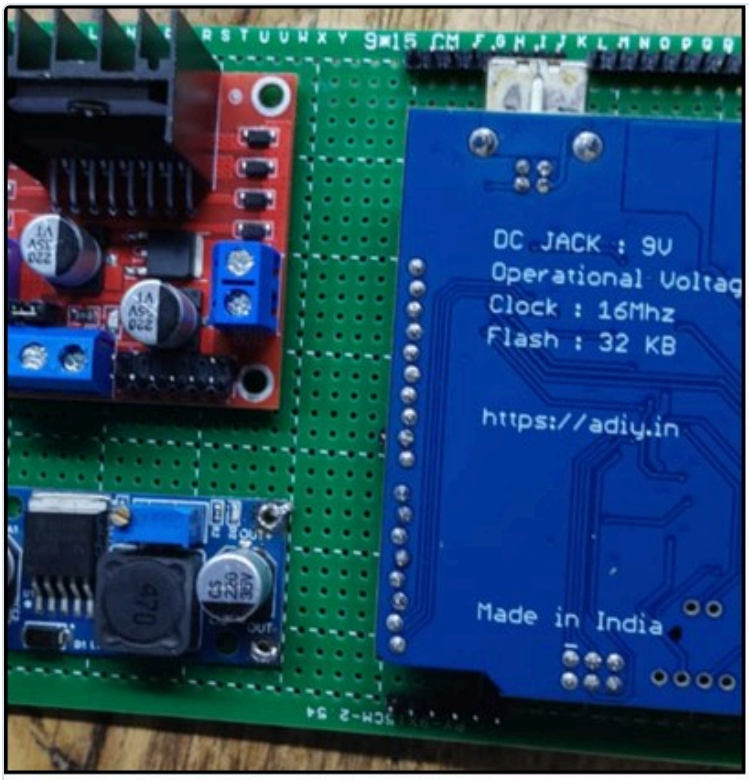
Test the controller, the ultrasonic sensors, the motor driver, the motors and the buck converter one by one before connecting them together. Check that both channels of the L298N get equal voltage with a multimeter, and keep the battery disconnected while you probe the circuit. A short during this kind of testing is what damaged our ESP32-S3.

Step 3⇒ Solder the Perfboard

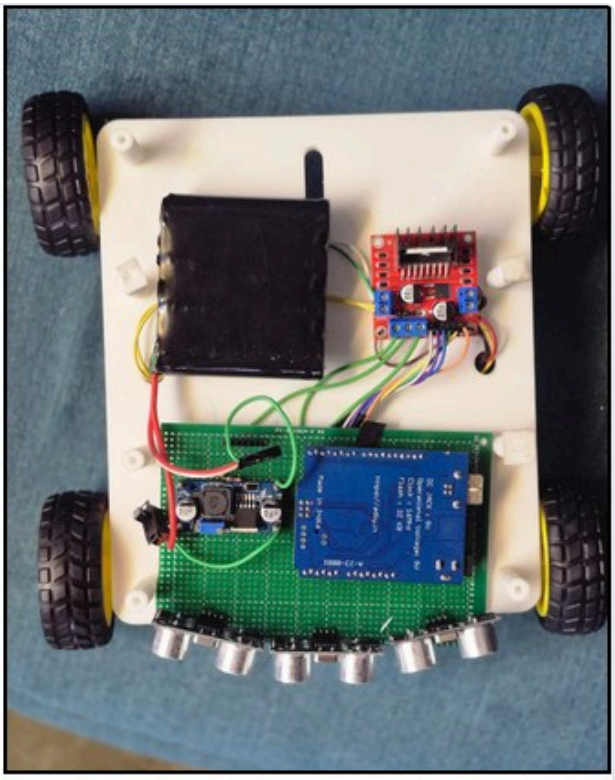
Move the circuit from the breadboard to a perfboard, with the L298N, the buck converter and the Uno on one board and a common ground. Every joint has to be soldered and checked, with a lot of wires in a small space.

Step 4⇒ Fit Everything on the Printed Base

Mount the four BO motors and the 65 mm wheels on the lower deck, and fix the perfboard and the battery in place. Then power the board from the 14.8V pack and check the Bluetooth link, the motor driver and the motors one after the other. Below are the perfboard and the base it went into.



STEP 3
L298N, BUCK CONVERTER, UNO ON PERFBOARD



STEP 4
FITTED ON THE PRINTED BASE

Step 5⇒ Mount the Three Ultrasonic Sensors at the Front

Put all three HC-SR04 sensors at the front of the lower deck, one at the centre and two at about +25° and −25°. With one sensor each at the front, left and right, our rover did not respond reliably. To try the ultrasonic side on its own first, refer to our HC-SR04 with ESP32 guide.

Setting Up the Three Controllers

MARS has three programs, one for each controller, and none of them is in the repository yet **TODO: add the Uno, Raspberry Pi and Nano code to the code folder**. Until they are published, write each one to the behaviour below.

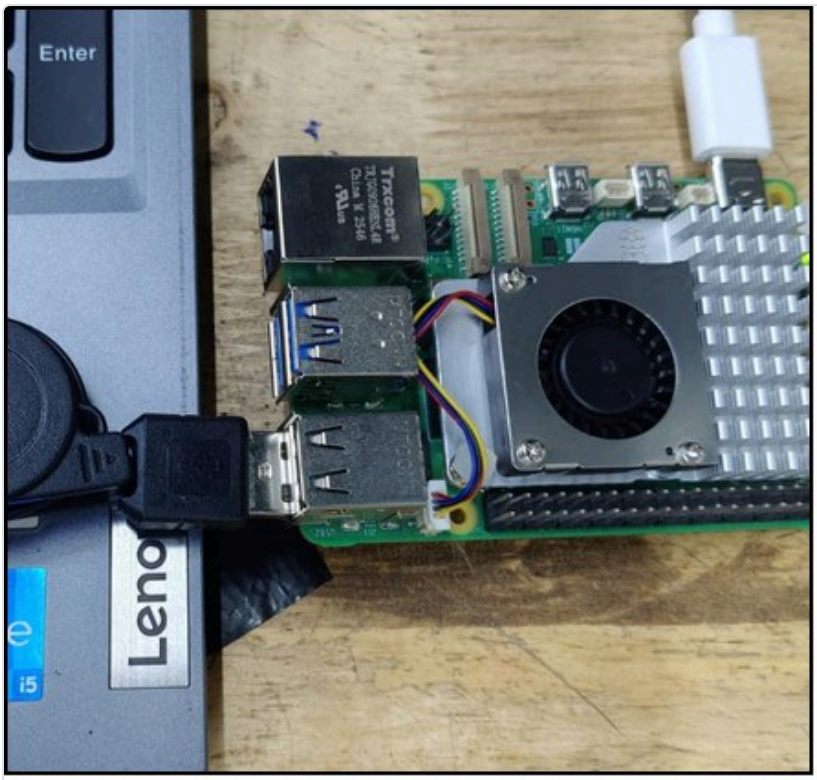
On the Arduino Uno, the program reads commands from the Bluetooth serial terminal app for forward, backward, left and right, reads the three HC-SR04 sensors on every pass, and stops the motors whenever an obstacle is in front. Pair the phone with the HC-05, open the serial terminal app and send the commands from there. Circuit Digest has a DIY Arduino Bluetooth car controlled by a mobile app with a similar HC-05 and L298N setup.

On the Raspberry Pi 5, install the OS, join the Pi to your network and connect the USB webcam. The Pi serves the webcam's video on a browser page on port 5000, which a phone on the same network can open.

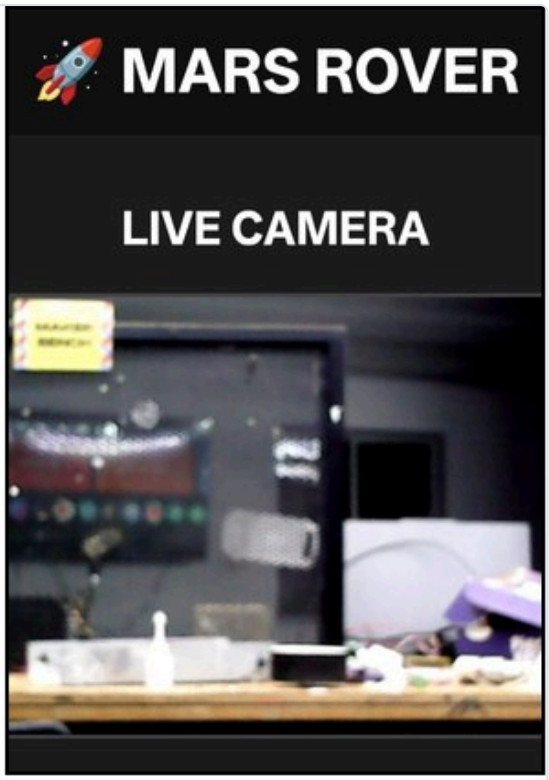
On the Arduino Nano, the program reads the gas sensor, the DHT11 and the soil-moisture probe, and drives the servo that raises and lowers the probe. For how a servo is driven from an Arduino, refer to our Arduino servo motor control tutorial.

Building the Application Modules

Each module mounts on the perforated upper deck with its own controller. Below are the Raspberry Pi 5 with its USB webcam and the MARS ROVER page it serves.



RASPBERRY PI 5 + USB WEBCAM
CAMERA MODULE



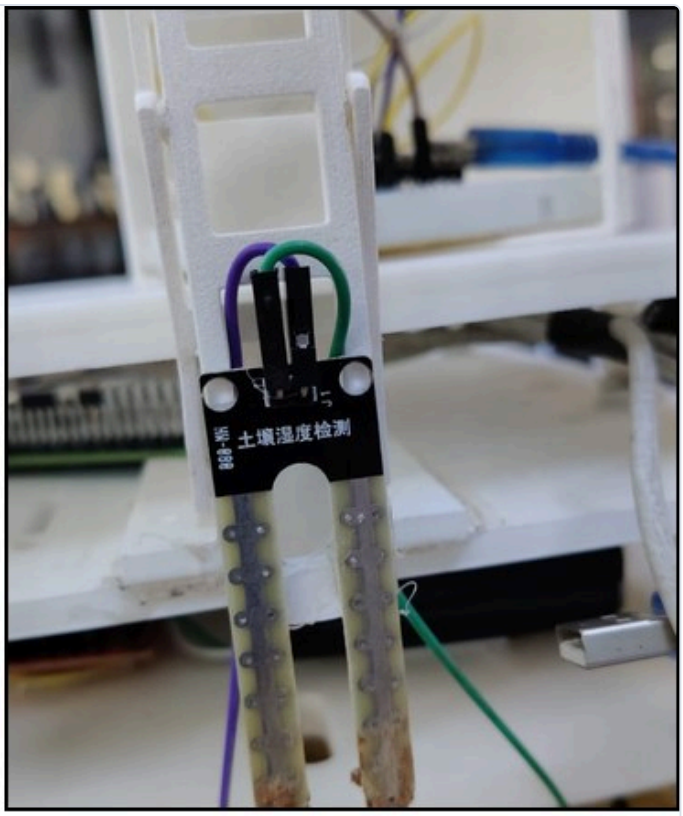
MARS ROVER PAGE
LIVE CAMERA ON PORT 5000

The page streams the webcam's video from port 5000. It also has rover control buttons, but the build logs only describe the video stream, so how those buttons reach the base is **TODO: link between the camera page and the base rover**. For another Pi and webcam build, do check out our Raspberry Pi surveillance robotic car as well.

The second module is the environmental module. Below are its sensors and the servo-driven lift for the soil probe.



MQ GAS SENSOR + DHT11
ON THE ARDUINO NANO MODULE

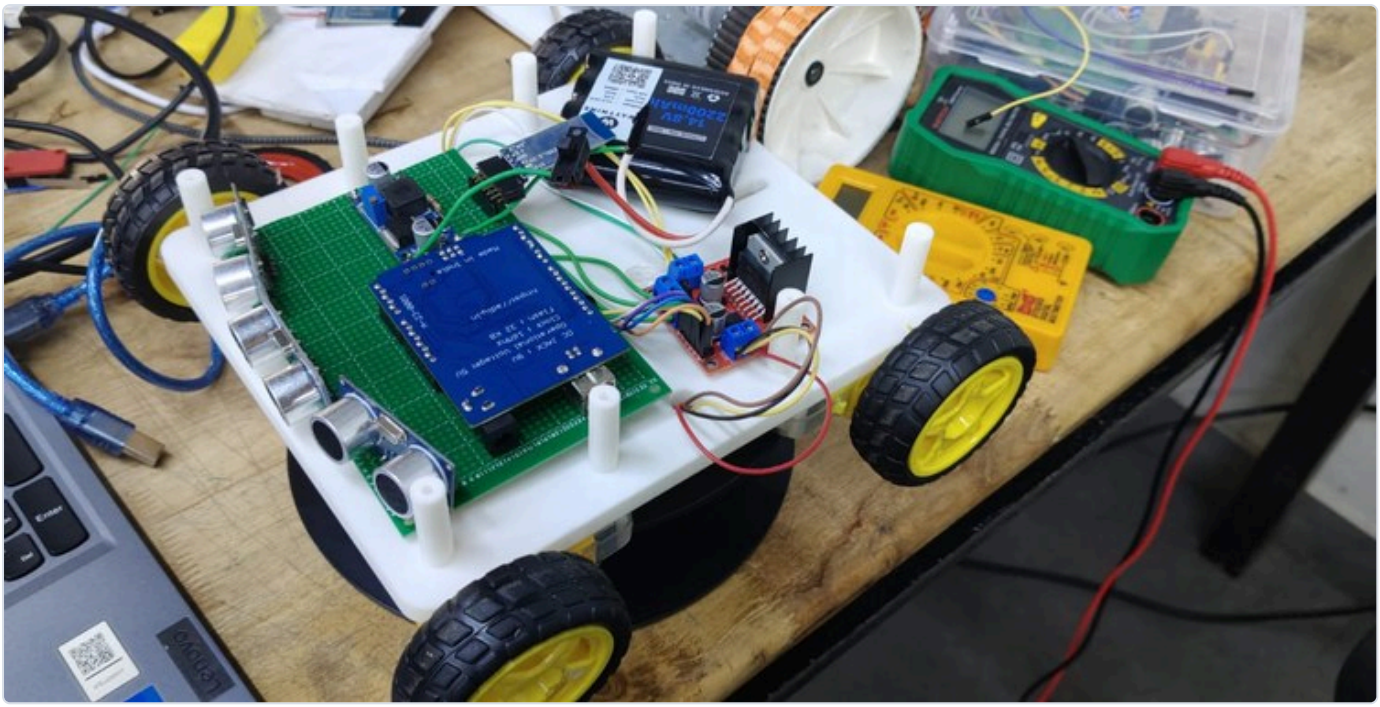


SOIL-MOISTURE PROBE
ON THE SERVO-DRIVEN LIFT

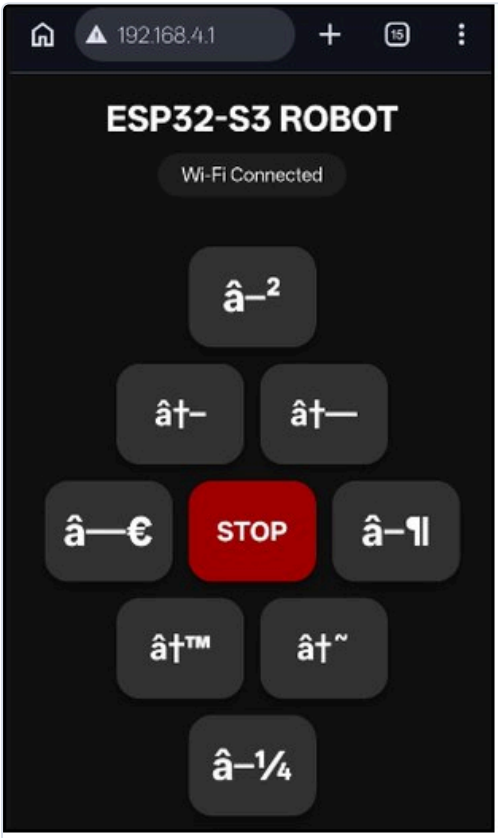
The Arduino Nano reads the gas sensor, the DHT11 and the soil-moisture probe. The probe sits on the printed ladder, and the servo raises and lowers it, so the probe goes into the soil only when a reading is needed. The air quality monitoring system on Circuit Digest reads an MQ gas sensor and a DHT11 with an Arduino in a similar way.

Testing and Calibrating the MARS Modular Rover

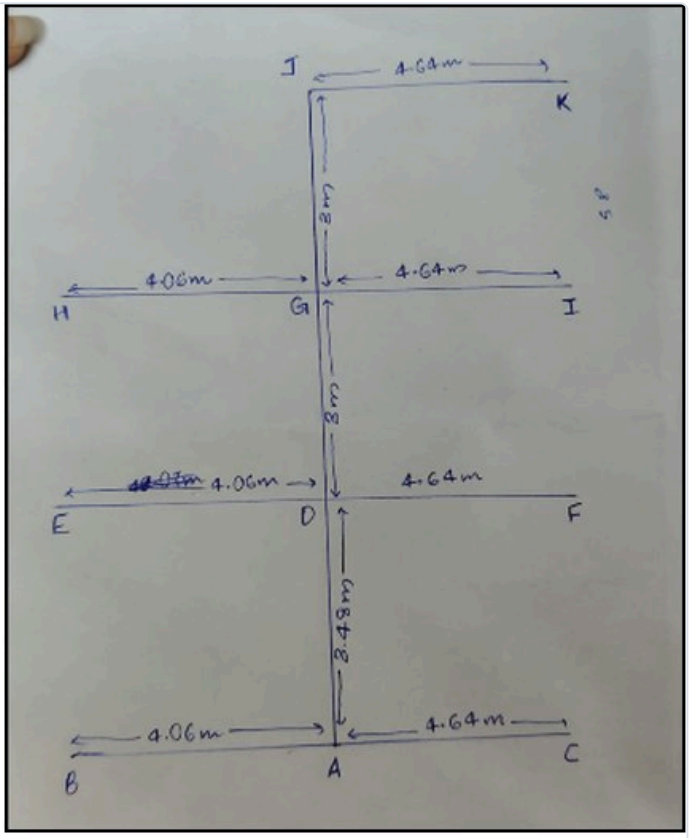
Test the ultrasonic sensors in a real room before you rely on them. Below is the rover during that testing.



Place the rover at different spots and distances from walls and objects, and compare the readings. This also shows up small problems with the wiring, the sensor positions and the component placement, which are easier to fix at this stage. For point-to-point runs, measure the room and draw a map. Below is ours, next to the Wi-Fi control page from our first ESP32-S3 controller.

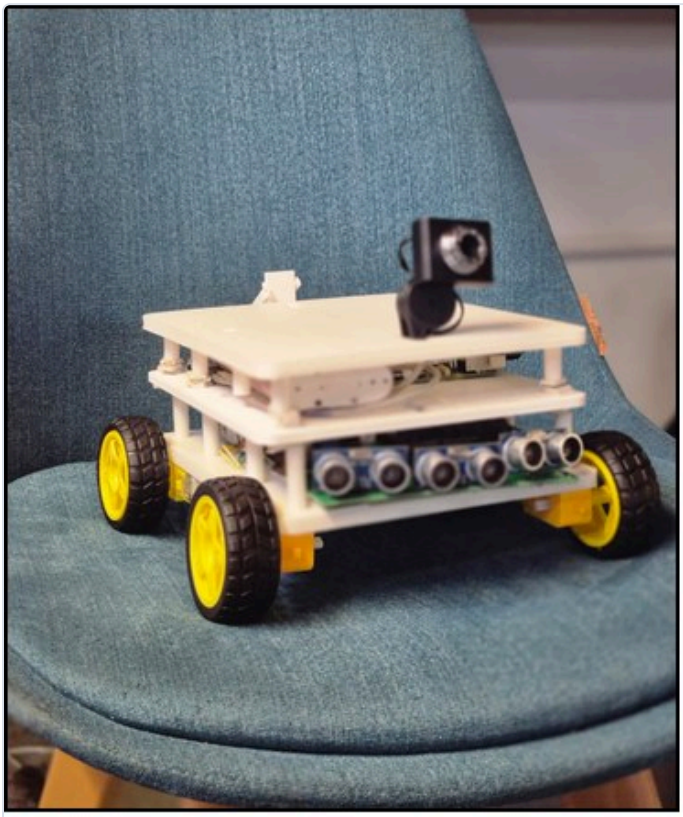


FIRST CONTROLLER
ESP32-S3 WI-FI CONTROL PAGE



ROOM MAP
POINTS A TO K FOR A-TO-B TESTS

The map has points A to K along a central line, with branches of 4.06 m on one side and 4.64 m on the other. We gave the Uno program the room dimensions and layout from this map, ran commands such as A to B, and adjusted the movement several times to get the path and the final position right. Then calibrate the sensors and the movement in a loop of testing, finding the error, adjusting and testing again. Check the camera module for streaming, and the environmental module for its sensors and the servo lift. Below is MARS with both modules on it.



FRONT
WEBCAM ON THE UPPER DECK



SIDE
SOIL PROBE LOWERED

The webcam sits on the upper deck, the three ultrasonic sensors look out from the front of the lower deck, and the soil probe hangs down on its lift at the side. The build logs record no sensor readings or test numbers TODO: sample readings and A-to-B accuracy. Mapping and fully autonomous navigation were planned as the next stage and were not reached.

Live Video Demonstration

Below is the video demonstration of MARS.

Watch the MARS modular autonomous rover system
YouTube · video not published yet (VIDEO_ID)

TODO: video of MARS driving on Bluetooth, stopping for an obstacle, streaming from the camera module and lowering the soil probe

Troubleshooting the MARS Rover

Symptom	Likely Cause	Fix
Controller dies when the 14.8V pack is connected	A short while testing with the battery connected	Disconnect the battery before probing, and check the board for shorts before powering it
The two L298N channels get different voltages	Not recorded in the logs	Measure both channels with a multimeter before the controller is connected
Obstacles at the sides are missed or the rover reacts late	Sensors spread to the front, left and right	Put all three at the front, at the centre and about ±25°
Wheels do not sit on the motor shafts	Wheel and shaft mismatch	Use wheels made for the motors, such as the 65 mm Universal Hub tyres
Printed parts do not fit the components	CAD size differs from the printed size	Measure the real parts, correct the CAD and allow for print tolerance
Bluetooth control does not work	Wrong connections, wrong pairing or serial settings	Check the HC-05 wiring, pair with the right device and check the serial settings and commands
Ultrasonic readings are wrong	Wiring, supply or a blocked path	Check the wiring and supply, give each sensor a clear path and check the distance logic
Environmental readings are unstable	Loose connections or sensors not yet settled	Check the connections and supply voltage, let the sensors stabilise and calibrate them

Conclusion

With these steps, you can build a MARS base that drives on Bluetooth commands and stops for obstacles, a Raspberry Pi 5 module that streams live video and an Arduino Nano module that reads its sensors and lowers the soil probe. The modules work, but each one still runs on its own controller. There is still room for improvement, such as a standard module mount with power and data connections, a link between the modules and the base, battery monitoring, and the mapping and autonomous navigation that were planned next. Further, one can make the base avoid obstacles instead of only stopping, just like this obstacle avoiding robot using Arduino and ultrasonic sensor on Circuit Digest.

MARS GitHub Repository

The MARS repository has the README, the build logs with their photos, and five STL files: the lower deck, the upper deck and three parts of the soil probe lift. The code folder only has a README for now. MARS was built at Tinkerspace by Riz Mariya Davis as part of the Builder in Residence programme, and the repository is released under the GNU GPL v3 licence.

Code & Schematics (GitHub) · Download Zip

Frequently Asked Questions About the MARS Modular Rover

⇥ 1. Why does MARS use an Arduino Uno and not an ESP32?

The first controller was an ESP32-S3, and it was damaged by a short circuit while the motor driver was being tested with the 14.8V pack connected. The Uno replaced it, and since the Uno has no Wi-Fi, the rover moved to Bluetooth control with an HC-05.

⇥ 2. Why are all three ultrasonic sensors at the front?

With sensors at the front, left and right, the rover did not respond reliably. Placing all three at the front, at the centre and about ±25°, gave better coverage in the direction the rover moves.

⇥ 3. Can the modules be swapped while the rover is running?

Not yet. Each module has its own controller and sits on the perforated upper deck, but there is no standard mount or shared power and data connection, which is one of the planned improvements.

⇥ 4. Does MARS navigate on its own?

Partly. It stops for obstacles and was tested on A-to-B runs using a measured room map, with the movement adjusted by hand several times. Mapping and full autonomous navigation were not reached.

Rover and Robotics Projects

Explore more rover builds, from Bluetooth driving to a Pi camera on wheels and an obstacle-avoiding robot.

Build your own Mars Rover Robot using Arduino:
The planet Mars has captivated our imagination for centuries, and this build makes a small Mars rover of our own with an Arduino.

DIY Web Controlled Raspberry Pi Surveillance Robotic Car:
In this DIY session we are building a web controlled surveillance robotic car using raspberry pi and a webcam.

Obstacle Avoiding Robot using Arduino and Ultrasonic Sensor:
An Arduino obstacle avoiding robot is a smart, autonomous robot that detects obstacles on its path using sensors and then navigates around the obstacle by changing direction.

Complete Project Code

The code for MARS is not in the repository yet. The code folder holds only a README, so there is no firmware to list here **TODO: add the complete code once it is published**.